

Welcome to ISA 8-Bit Ethernet Card!

Documentation, including the detailed assembly instructions, specifications, design files, and the software can be found at https://github.com/skiselev/isa8_eth



Installation Instructions:

1. Use provided M3 screws to assemble the 3D printed ISA bracket to the card. The bracket mount tabs should be on the bottom side of the card, and the screws are screwed from the top side
2. Set card switches:
 - If using switches to configure the card, set switches according to the tables below. Make sure that configured resources - I/O port addresses, IRQ, memory window for the boot ROM - are available
 - If using jumperless mode, set switches SW1.1 and SW1.2 to OFF position
 - If using PnP mode, set switch SW1.1 is to OFF position and SW1.2 to in ON position**Note: The card is shipped configured in jumperless mode. The default configuration is:
I/O address: 0x260, IRQ: 2, Boot ROM: Disabled**
3. Install card in an available ISA slot of your computer
4. If needed, use RSET8019 utility to change card configuration in jumperless mode
5. Install the software. Use the patched 8-bit NE2000 packet driver, provided in the GitHub repository linked above
 - The packet driver requires the packet driver interrupt number (usually 0x60), the IRQ number and the I/O address arguments. For example: NE2000 0x60 2 0x260

Configuration Switches

SW1 - CONFIG

This switch block sets the configuration mode

Configuration Mode	Switch Position
Jumperless / RSET8019	SW1.1 = OFF, SW1.2 = OFF
Switches	SW1.1 = ON, SW1.2 = OFF
PnP	SW1.1 = OFF, SW1.2 = ON

SW2 - IRQ

This switch block, when the card is in the **Switches** configuration mode, sets the Ethernet controller IRQ

IRQ	Switch Position
IRQ2 (on PC/XT) or IRQ9 (on AT)	SW2.1 = OFF, SW2.2 = OFF
IRQ3	SW2.1 = ON, SW2.2 = OFF
IRQ4	SW2.1 = OFF, SW2.2 = ON
IRQ5	SW2.1 = ON, SW2.2 = ON

SW3 - IO_ADDR

This switch block, when the card is in the **Switches** configuration mode, sets the Ethernet controller I/O address

I/O Address	SW3.1	SW3.2	SW3.3	SW3.4
0x200	OFF	OFF	ON	OFF
0x220	ON	OFF	ON	OFF
0x240	OFF	ON	ON	OFF
0x260	ON	ON	ON	OFF
0x280	OFF	OFF	ON	ON
0x2A0	ON	OFF	ON	ON
0x2C0	OFF	ON	ON	ON
0x2E0	ON	ON	ON	ON
0x300	OFF	OFF	OFF	OFF
0x320	ON	OFF	OFF	OFF
0x340	OFF	ON	OFF	OFF
0x360	ON	ON	OFF	OFF
0x380	OFF	OFF	OFF	ON
0x3A0	ON	OFF	OFF	ON
0x3C0	OFF	ON	OFF	ON
0x3E0	ON	ON	OFF	ON

SW4 - ROM_ADDR

This switch block, when the card is in the **Switches** configuration mode, sets the Ethernet controller boot ROM address and size

ROM Address	ROM Size	SW4.1	SW4.2	SW4.3	SW4.4	SW4.5
ROM Disabled	Not applicable	*	*	*	OFF	OFF
0xC0000	16 KiB	OFF	OFF	OFF	OFF	ON
0xC4000	16 KiB	ON	OFF	OFF	OFF	ON
0xC8000	16 KiB	OFF	ON	OFF	OFF	ON
0xCC000	16 KiB	ON	ON	OFF	OFF	ON
0xD0000	16 KiB	OFF	OFF	ON	OFF	ON
0xD4000	16 KiB	ON	OFF	ON	OFF	ON
0xD8000	16 KiB	OFF	ON	ON	OFF	ON
0xDC000	16 KiB	ON	ON	ON	OFF	ON
0xC0000	32 KiB	OFF	OFF	OFF	ON	OFF
0xC8000	32 KiB	ON	OFF	OFF	ON	OFF
0xD0000	32 KiB	OFF	ON	OFF	ON	OFF
0xD8000	32 KiB	ON	ON	OFF	ON	OFF
0xC0000	64 KiB	OFF	OFF	ON	ON	OFF
0xD0000	64 KiB	ON	OFF	ON	ON	OFF
0xC0000	Page mode	OFF	OFF	OFF	ON	ON
0xC4000	Page mode	ON	OFF	OFF	ON	ON
0xC8000	Page mode	OFF	ON	OFF	ON	ON
0xCC000	Page mode	ON	ON	OFF	ON	ON
0xD0000	Page mode	OFF	OFF	ON	ON	ON
0xD4000	Page mode	ON	OFF	ON	ON	ON
0xD8000	Page mode	OFF	ON	ON	ON	ON
0xDC000	Page mode	ON	ON	ON	ON	ON

